11. EXPLORE TOR NETWORK FOR ANONYMOUS COMMUNICATION

DATE: 01/04/2026

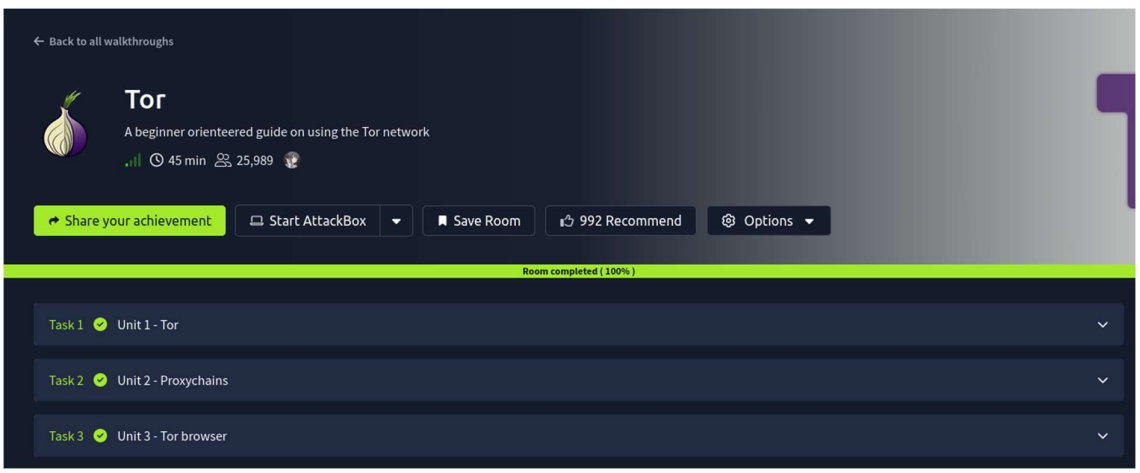
Aim:

To understand and use the Tor network for secure and anonymous communication.

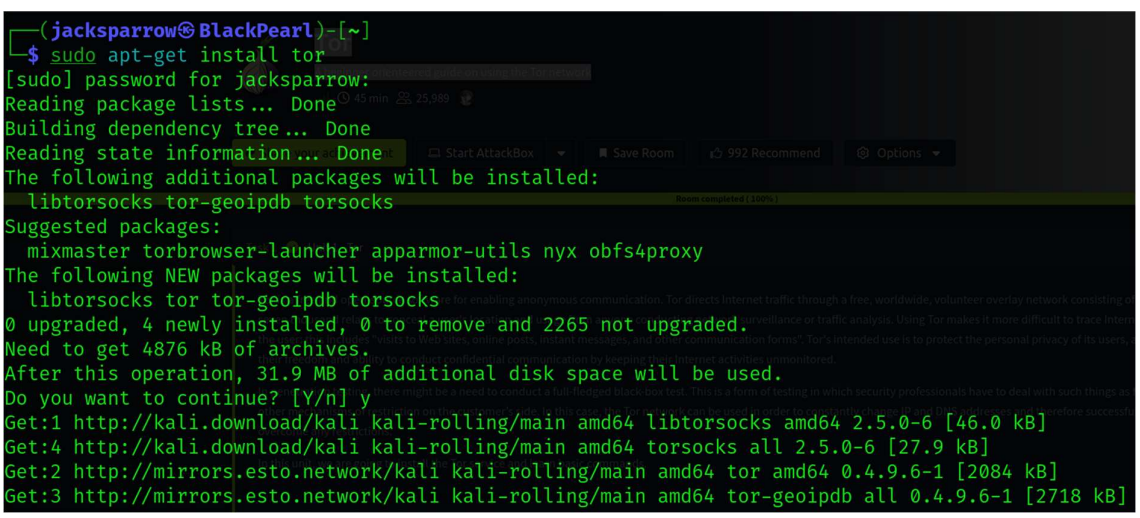
Procedure:

1. Log into TryHackMe and open the Tor room.
2. Study the basics of the Tor network and its purpose.
3. Understand how traffic is routed through multiple nodes for anonymity.
4. Install and launch the Tor Browser.
5. Explore secure browsing and verify anonymity using test methods.
6. Learn limitations and risks associated with Tor usage.
7. Complete the tasks and confirm completion.

Tasks:



TASK 1 – Unit 1-Tor



```
┌──(jacksparrow㉿BlackPearl)-[~]
└─$ sudo apt-get install tor
[sudo] password for jacksparrow:
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
The following additional packages will be installed:
  libtorsocks tor-geoipdb torsocks
Suggested packages:
  mixmaster torbrowser-launcher apparmor-utils nyx obfs4proxy
The following NEW packages will be installed:
  libtorsocks tor tor-geoipdb torsocks
0 upgraded, 4 newly installed, 0 to remove and 2265 not upgraded.
Need to get 4876 kB of archives.
After this operation, 31.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 libtorsocks amd64 2.5.0-6 [46.0 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 torsocks all 2.5.0-6 [27.9 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 tor amd64 0.4.9.6-1 [2084 kB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 tor-geoipdb all 0.4.9.6-1 [2718 kB]
```

```
┌──(jacksparrow㉿BlackPearl)-[~]
└─$ sudo su
┌──(root㉿BlackPearl)-[/home/jacksparrow]
└─# service tor start

┌──(root㉿BlackPearl)-[/home/jacksparrow]
└─# service tor status
● tor.service - Anonymizing overlay network for TCP (multi-instance-master)
     Loaded: loaded (/usr/lib/systemd/system/tor.service; disabled; preset: disabled)
     Active: active (exited) since Mon 2026-04-20 12:42:05 IST; 6min ago
 Invocation: de39ba8be31945ca999fea927fdb6330
    Process: 12246 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 12246 (code=exited, status=0/SUCCESS)

Apr 20 12:42:05 BlackPearl systemd[1]: Starting tor.service - Anonymizing overlay network for TCP (m>
Apr 20 12:42:05 BlackPearl systemd[1]: Finished tor.service - Anonymizing overlay network for TCP (m>
lines 1-9/9 (END)
```

```
┌──(root㉿BlackPearl)-[/home/jacksparrow]
└─# service tor stop
```

Answer the questions below

Run `apt-get install tor` to install/update your Tor packages.

No answer needed — ✓ Correct Answer

Run `service tor start` to start the Tor service.

No answer needed — ✓ Correct Answer

Run `service tor status` to check Tor's availability.

No answer needed — ✓ Correct Answer

Run `service tor stop` to stop the Tor service.

No answer needed — ✓ Correct Answer

TASK 2 – Unit 2-ProxyChains

```
┌──(root㉿BlackPearl)-[/home/jacksparrow]
└─# apt install proxychains
Installing:
  proxychains

Installing dependencies:
  libproxychains3

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2265
  Download size: 22.7 kB
  Space needed: 74.8 kB / 32.6 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libproxychains3 amd64 3.1-9+b3 [13.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 proxychains all 3.1-9 [9140 B]
Fetched 22.7 kB in 11s (2038 B/s)
Selecting previously unselected package libproxychains3:amd64.
(Reading database ... 395381 files and directories currently installed.)
Preparing to unpack .../libproxychains3_3.1-9+b3_amd64.deb ...
```

```
root@BlackPearl: /home/jacksparrow
File  Actions  Edit  View  Help
  GNU nano 8.1                    /etc/proxychains.conf *
# proxychains.conf  VER 3.1
#
#        HTTP, SOCKS4, SOCKS5 tunneling proxifier with DNS.
#

# The option below identifies how the ProxyList is treated.
# only one option should be uncommented at time,
# otherwise the last appearing option will be accepted
#
dynamic_chain
#
# Dynamic - Each connection will be done via chained proxies
# all proxies chained in the order as they appear in the list
# at least one proxy must be online to play in chain
# (dead proxies are skipped)
# otherwise EINTR is returned to the app
```

```
┌──(jacksparrow㉿BlackPearl)-[~]
└─$ proxychains firefox
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
```

Answer the questions below

Let's start with running `apt install proxychains` to install/update proxychains tool.

No answer needed — ✓ Correct Answer

Now it's time to configure proxychains to work properly.

Run `nano /etc/proxychains.conf` to edit the settings. (Note: You can use any text editing tool instead of nano.)

No answer needed — ✓ Correct Answer

We can now see, that most of the methods are under comment mark. You can read their description and decide on using one of them in the future. For this lesson let's uncomment `dynamic_chain` and comment others (simply put `#` to the left). Additionally, it is useful to uncomment `proxy_dns` in order to prevent DNS leak. Scroll through the document and see whenever you want to add some additional proxies at the bottom of the page (which is not required at this point).

Apply all the settings.

No answer needed — ✓ Correct Answer

Now let's check our settings.

Start the TOR service and run `proxychains firefox`. Usually, you are required to put `proxychains` command before anything in order to force it to transfer data through Tor.

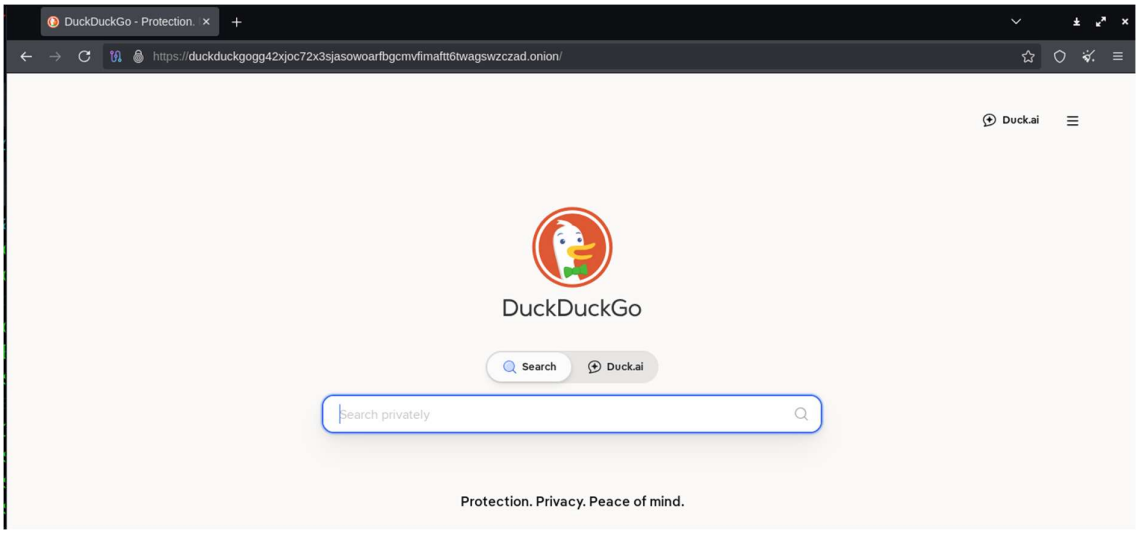
No answer needed — ✓ Correct Answer

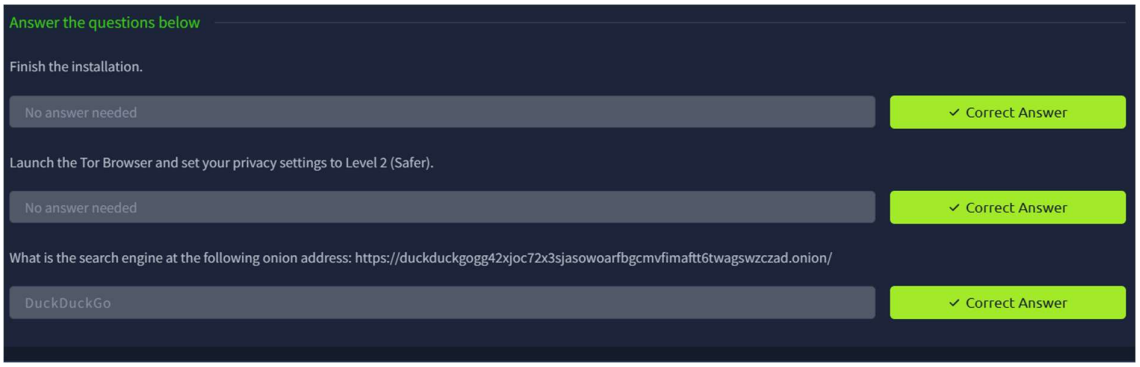
After the Firefox has loaded, check if your IP address has changed with any website that provides such information. Also, try running a test on `dnsleaktest.com` and see if your DNS address changed too.

NOTE: All other web browser windows should be closed before opening Firefox through proxychains!

No answer needed — ✓ Correct Answer

TASK 3 – Unit 3-Tor Browser





Result:

Thus the TryHackMe room “Tor” was completed successfully.